Returns to Schooling in Brazil, 2012–2025: Design-Based Mincer Estimates from the PNAD Contínua

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Abstract

Using the annual first-visit microdata of Brazil's PNAD Contínua for 2012–2019 and 2022–2025, this paper estimates Mincer earnings equations for 1.77 million occupied workers aged 25–64 with inference that respects the survey's strata, primary sampling units and weights. It replicates, for Brazil, the OLS benchmark that Duflo (2001) reports before her instrumental-variables estimates of the return to schooling. The pooled return to a year of schooling is 9.8 percent and the year-by-year estimates stay between 9.3 and 10.2 percent, with the lowest values in 2022–2024. That stability hides a strong compression of premia by completed level: relative to no schooling, the premium for a completed degree fell from 1.62 to 1.29 log points between 2012 and 2025. The schooling–earnings profile is convex, with an implied 20 percent per year between a completed secondary and a completed tertiary education; returns are higher for women than for men, lower for Black and Brown workers, and highest in the Nordeste. Ignoring the survey design understates standard errors by a factor of three. All estimates are descriptive; the paper states what a causal design for Brazil would require. Code and data pipeline reproduce every number from IBGE's public files.

Resumo. Com os microdados anuais da primeira visita da PNAD Contínua para 2012–2019 e 2022–2025, este artigo estima equações de Mincer para 1,77 milhão de ocupados de 25 a 64 anos, com inferência que respeita os estratos, as unidades primárias de amostragem e os pesos da pesquisa. Trata-se de uma replicação, para o Brasil, do retorno por MQO que Duflo (2001) reporta antes de suas estimativas por variáveis instrumentais. O retorno a um ano de estudo é de 9,8% no painel empilhado e fica entre 9,3% e 10,2% nas estimativas ano a ano, com os menores valores em 2022–2024. Essa estabilidade esconde uma forte compressão dos prêmios por nível concluído: em relação a quem não tem instrução, o prêmio do ensino superior completo caiu de 1,62 para 1,29 pontos de log entre 2012 e 2025. O perfil de rendimentos é convexo, com cerca de 20% ao ano entre o ensino médio e o superior completos; os retornos são maiores para mulheres do que para homens, menores para pretos e pardos e mais altos no Nordeste. Ignorar o desenho amostral subestima os erros-padrão por um fator de três. As estimativas são descritivas; o artigo explicita o que um desenho causal para o Brasil exigiria.

JEL: I26, J31, O15, O54. **Keywords:** returns to schooling; Mincer equation; Brazil; PNAD Contínua; complex survey design; replication.

1 Introduction

How much more do Brazilian workers earn for each additional year of schooling, and has that premium changed since 2012? The question is old, but the answer matters for current policy: Brazil expanded schooling faster than almost any large country after the 1990s, and the payoff to that expansion, together with its distribution by sex, race and region, shapes both the private incentive to stay in school and the inequality that education is expected to reduce. The question also sits at the centre of the empirical development literature that this paper series follows. In the Week 3 paper of the MIT DEDP course *Foundations of Development Policy*, Duflo (2001) uses a school-construction program in Indonesia to estimate the causal return to schooling and compares it with the ordinary least-squares (OLS) Mincer return of 7.8 percent that she estimates in the same data. This paper estimates that benchmark for Brazil.

The contribution is deliberately modest and transparent. Using the annual first-visit microdata of the *Pesquisa Nacional por Amostra de Domicílios Contínua* (PNAD Contínua) for 12 survey years between 2012 and 2025, I estimate Mincer earnings equations for 1,773,567 occupied workers aged 25–64, with inference that respects the survey's stratified, clustered and weighted design. Every estimate uses the strata, primary sampling units and first-visit weights published by IBGE, the Brazilian statistical office, so that standard errors reflect how the sample was actually drawn (Lumley 2004, 2010). Before any regression is run, the pipeline reproduces IBGE's own published aggregates from the same microdata, which checks that weights, deflators and sample definitions are applied correctly. All code downloads the data and regenerates every number in this paper from a clean copy of the repository.

The evidence is descriptive. A Mincer coefficient estimated by OLS does not identify the causal effect of schooling on earnings: unobserved ability and family background raise both schooling and earnings, and the sample of people with earnings is selected (Card 1999; Heckman et al. 2006). Duflo's instrumental-variables estimates were designed to address exactly that problem; a later reanalysis by Roodman (2022) finds them weakly identified, which is a useful reminder that the causal return is hard to pin down even with a large natural experiment. What a careful descriptive estimate can deliver is a precise, honestly bounded measurement of the earnings–schooling gradient, how it moved over a decade of expansion, and where it differs across groups. That is what the paper reports, and it states plainly what a causal counterpart for Brazil would require.

Four results stand out. First, the pooled return to a year of schooling is 9.8% (design-based standard error 0.05%), and the year-by-year estimates stay within a narrow band, from 9.3% to 10.2%, with the lowest values after the pandemic. Second, that stability hides a strong compression of premia by completed level: relative to workers with no schooling, the premium for a completed degree fell from 1.62 log points in 2012 to 1.29 in 2025, and the premia for completed fundamental and secondary education fell by similar proportions. The two facts are consistent because the earnings–schooling profile is steeply convex, with an implied 20% per year between a completed secondary and a completed tertiary education against 5% per year of fundamental education, and the workforce moved up the profile as mean schooling rose from 9.8 to 11.6 years. Third, the gradient differs by group: it is 0.9% points higher for women than for

men and 2.5% points lower for Black and Brown workers than for White and Asian workers, so the conditional gender gap narrows with schooling while the racial gap widens; across regions it is highest in the Nordeste. Fourth, inference matters: unweighted OLS with heteroskedasticity-robust standard errors gives a standard error 3.4 times smaller than the design-based one, while clustering by primary sampling unit recovers most of the design variance.

The rest of the paper is organised as follows. Section 2 relates the exercise to the original paper and the lessons of its reanalysis. Section 3 describes the data, the sample and the official-aggregate checks. Section 4 sets out the specifications and the inference. Section 5 presents the results, Section 6 the robustness checks, Section 7 the limitations, and Section 8 concludes.

2 Relation to the original paper

Duflo (2001) studies the INPRES program, under which Indonesia built 61,807 primary schools between 1973–74 and 1978–79, about two per 1,000 children aged 5–14 in 1971. Exposure depends on two things an adult cannot choose: year of birth, because children aged 12 or more in 1974 had already left primary school, and district of birth, because the allocation rule sent more schools to districts with low enrolment in 1972. Combining the two in a difference-in-differences design with region and cohort fixed effects, she estimates that each school per 1,000 children raised schooling by 0.12–0.19 years and wages by 1.5–2.7 percent for fully exposed cohorts. Using the cohort-by-intensity interactions as instruments, the two-stage least-squares return to a year of schooling ranges from 6.8 to 10.6 percent, close to the OLS estimate of 7.8 percent in the same sample of wage earners (her Table 7). She concludes that, in that context, OLS returns are not biased upward by omitted family background, and notes that her OLS return is below earlier Indonesian estimates, consistent with returns falling as schooling expands.

Roodman (2022) reconstructs the data and re-runs the analysis. Four of his changes are, in his words, relatively uncontroversial: correcting data-entry errors in the baseline variables, clustering standard errors by the geographic unit of treatment, following the cohorts into later surveys, and using inference that is robust to weak identification. He also documents that the original code applied survey weights in the two-by-two comparisons but not in the regressions, a choice the paper did not discuss. The headline 1995 estimates are generally not robust to these revisions and the returns to schooling are weakly identified, although the pattern evidence, with effects appearing only in the exposed cohorts and at the expected grade levels, partly persists.

Three lessons from that exchange shape this paper. First, inference choices are stated, not left to the code: weights, strata, primary sampling units and the resulting degrees of freedom are reported, and the main estimate is re-run under the common shortcuts of ignoring the design or using cluster-robust standard errors alone. Second, controls are fixed in advance and are the same in every table. Third, pattern evidence is reported alongside the single coefficient: premia by completed level, which reveal sheepskin effects, and year-by-year estimates, which show the trend that Duflo reads into her lower OLS return.

What the paper does not do is replicate Duflo's identification. Brazil has no school-construction

program with a documented allocation rule and cohort-by-region exposure on the scale of INPRES; the closest analogue, the 1998 FUNDEF reform of school finance, redistributed funds across municipalities with a mechanical rule, but studying it requires region of birth, which the PNAD Contínua does not record (only the state of residence), and many treatment clusters, which means municipalities in the Census rather than the 27 states. That design is a separate paper. This one estimates, carefully, the OLS benchmark against which such a design would be judged.

3 Data

3.1 The PNAD Contínua

The PNAD Contínua is IBGE's continuous household survey, in the field since 2012. It samples about 211,000 households per quarter in a stratified, two-stage design: primary sampling units (UPAs, census enumeration areas) are selected within strata defined by state, metropolitan status and other criteria, and households are selected within UPAs. Households are interviewed five times in five consecutive quarters. IBGE publishes the first interview of each household as an annual "first-visit" file, which carries the full basic questionnaire on education and labour, and uses that file for its annual labour-income releases (IBGE 2024). This paper uses the first-visit files for every year released: 2012–2019 and 2022–2025. IBGE did not release first-visit microdata for 2020 and 2021, when fieldwork moved to the telephone during the pandemic, so those two years are absent from every figure and table. The data were downloaded on 11 October 2026 with the PNADcIBGE package (Braga and Assunção 2024), which also attaches IBGE's deflators.

Earnings are expressed at 2025 average prices using the deflators IBGE publishes for the first-visit files, with the series for habitual earnings applied to habitual earnings and the series for effective earnings applied to effective earnings. The design variables are the stratum, the UPA and the first-visit weight, which IBGE calibrates to population projections by post-stratum.

3.2 Sample and variables

The dependent variable is the log of real hourly earnings in the main job: habitual monthly earnings in the main job divided by weekly hours habitually worked in that job times 52/12. Years of schooling are IBGE's standardised measure, from 0 to 16, in which 9 years marks completed fundamental education, 12 completed *ensino médio* and 16 a completed degree; completed level is the companion categorical variable. Controls are age and its square, sex, race (White or Asian, Black or Brown, Indigenous, following IBGE's categories), urban residence and state fixed effects. The pooled specifications add year fixed effects.

The analysis sample is every occupied person aged 25–64 with positive habitual earnings and between 1 and 120 weekly hours in the main job, with schooling and race reported. Unlike Duflo's Indonesian survey, which recorded wages only for wage employees, the PNAD Contínua records earnings for the self-employed and for employers, so the main sample includes them; wage employees only and formal workers only are reported as robustness checks. Table 7 in the

appendix gives the unweighted count at each filter step and year; between 95.8% and 98.4% of occupied 25–64-year-olds enter the sample, the main loss being workers without earnings (unpaid family workers) in the earlier years. The final sample has 1,773,567 person-years.

Table 1 reports design-based means by year. Mean schooling in the sample rose from 9.84 years in 2012 to 11.59 in 2025, a gain of almost two years in little more than a decade; real hourly earnings at 2025 prices moved from R$ 19.62 to R$ 22.29, falling during the 2015–2016 recession and recovering after 2022. The share of women rose from 41.4% to 43.7% and the share of Black and Brown workers from 50.1% to 54.7%, while the share in formal work stayed near 64%.

Table 1: Analysis sample by year: design-based means

Year	Observations	Schooling	Hourly earnings	Age	Women	Black or Brown	Formal
2012	146,662	9.84	19.62	40.1	0.414	0.501	0.632
2013	152,993	9.96	20.28	40.2	0.417	0.506	0.642
2014	156,830	10.12	20.72	40.4	0.419	0.508	0.654
2015	155,348	10.27	20.13	40.6	0.421	0.516	0.656
2016	154,607	10.42	19.92	40.7	0.424	0.522	0.649
2017	152,398	10.56	19.91	40.8	0.426	0.528	0.634
2018	151,572	10.78	20.09	41.0	0.430	0.535	0.626
2019	150,773	10.88	20.04	41.1	0.431	0.540	0.625
2022	130,659	11.19	19.60	41.3	0.431	0.539	0.623
2023	133,978	11.30	20.41	41.5	0.434	0.544	0.625
2024	137,271	11.40	21.06	41.6	0.436	0.553	0.625
2025	150,476	11.59	22.29	41.7	0.437	0.547	0.633

Occupied persons aged 25-64 with positive habitual earnings and 1-120 weekly hours in the main job. Means use the first-visit weights. Schooling in years; hourly earnings in the main job at 2025 prices (R$); women, Black or Brown and formal are shares. Formal: signed-card or statutory employees, or self-employed and employers contributing to social security. Weighted sample sizes range from 68.5 million (2012) to 83.7 million (2025) person-years.

3.3 Reproducing official aggregates

Because weights, deflators and domain definitions are where survey analyses most often go wrong, the pipeline reproduces two published IBGE series from the same design object before any model is estimated (Table 8 in the appendix). The mean real monthly labour income of occupied persons aged 14 or more, habitually and effectively received in all jobs at 2025 prices, is reproduced from SIDRA table 7445 to within 0.01 percent in every year except 2022, where the current microdata file gives values 3.3 percent above the published ones in both measures; the published 2022 values appear to predate the 2025 revision of the microdata, and the discrepancy is recorded rather than hidden. The mean years of schooling of persons aged 25 or more, in total and by sex, is reproduced from SIDRA table 7126 to within 1.0%; that series is published from

the second-quarter education supplement and rounded to one decimal, so a closer match is not expected. The effective-earnings check also documents a definitional detail: IBGE's published mean averages effective earnings, including zero months, over everyone with positive habitual earnings.

4 Method

4.1 Specifications

The baseline is the Mincer equation with a linear return to schooling,

$$\ln w_i = \beta\, S_i + \gamma_1 \text{age}_i + \gamma_2 \text{age}_i^2 + \mathbf{x}_i'\delta + \alpha_{s(i)} + \varepsilon_i, \tag{1}$$

where S_i is years of schooling, $\mathbf{x}_i$ holds the sex, race and urban indicators and $\alpha_{s(i)}$ is a state fixed effect. It is estimated separately for each survey year, which gives the trend in β, and pooled over all years with year fixed effects. A second specification replaces S_i by indicators for the completed level of education, which lets the return differ across the schooling distribution and exposes sheepskin effects at the completion thresholds. A third interacts S_i with sex and with race, so that the return can differ by group; the return for each sex-by-race group is a linear combination of coefficients with a standard error computed from the full covariance matrix. Regional estimates re-run Equation 1 within each macro-region. Age and its square stand in for potential experience; using experience defined as age minus schooling minus six would make the schooling coefficient depend on the experience adjustment, so the age polynomial is preferred.

4.2 Inference

All estimates are design-based (Lumley 2004). Each year's design object is built from the full first-visit file with the UPA as the sampling unit, the stratum as the stratum and the first-visit weight as the weight, and the analysis sample is then taken as a domain of that design, so that the variance estimator keeps the full count of sampling units per stratum. Strata with a single sampled UPA after subsetting are centred at the grand mean. Regression coefficients are estimated by survey-weighted least squares and their variances by the linearisation estimator of the `survey` package; the reference distribution is a t with degrees of freedom equal to the number of UPAs minus the number of strata, about 44,580 in the pooled design. The pooled design stacks all years, keeps the stratum codes and treats the same UPA in different years as one cluster, which is conservative for repeated areas of the master sample.

Two shortcuts are reported for comparison in the robustness table: unweighted OLS with heteroskedasticity-robust (HC1) standard errors, which ignores the design altogether, and weighted OLS with standard errors clustered by UPA, which uses the weights and the clustering but not the strata. The comparison shows what each layer of the design contributes (Solon et al. 2015; Abadie et al. 2023).

Throughout, a coefficient of β on years of schooling is read as a log-point difference; for small values it is close to a percentage difference in hourly earnings between workers who differ by one

year of schooling and share the same age, sex, race, area and state. It is not the causal effect of a year of schooling.

5 Results

5.1 The return to a year of schooling, 2012–2025

Table 3 reports the pooled specifications and Table 2 and Figure 1 the year-by-year estimates of Equation 1. Pooled over the twelve survey years with year fixed effects, a year of schooling is associated with 9.8% higher hourly earnings (standard error 0.05%, 1,773,567 observations). Year by year, the coefficient is 9.9% in 2012, drifts down to 9.7% in 2016, rises to 10.2% in 2019, and is lowest in the three post-pandemic years, 9.4%, 9.3% and 9.3%, before returning to 9.8% in 2025. With design-based standard errors of about 0.10% points, the differences between adjacent years are mostly within sampling error, but the gap between 2018–2019 and 2022–2024 is not: the confidence intervals [10.0%, 10.4%] and [9.1%, 9.5%] do not overlap. The Brazilian benchmark is therefore above the 7.8 percent that Duflo (2001) estimates by OLS for Indonesian wage earners in 1995 and in the upper part of the range that Psacharopoulos and Patrinos (2018) report for Latin America.

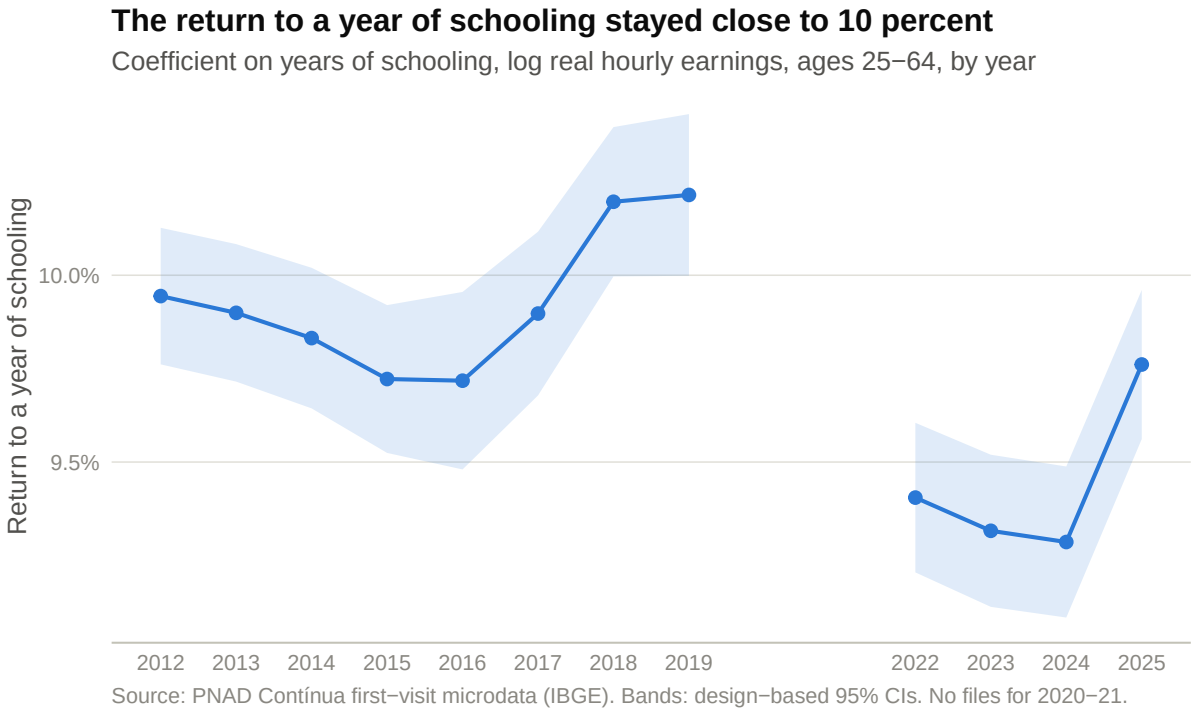


Figure 1: The return to a year of schooling by survey year. Coefficient on years of schooling in the linear specification with state fixed effects, one regression per year, workers aged 25–64. Shaded bands are design-based 95 percent confidence intervals. No first-visit microdata were released for 2020 and 2021.

Table 2: Return to a year of schooling by survey year

Year	Estimate	SE	95% CI	Observations
2012	0.0994	(0.0009)	[0.098, 0.101]	146,662
2013	0.0990	(0.0009)	[0.097, 0.101]	152,993
2014	0.0983	(0.0010)	[0.096, 0.100]	156,830
2015	0.0972	(0.0010)	[0.095, 0.099]	155,348
2016	0.0972	(0.0012)	[0.095, 0.100]	154,607
2017	0.0990	(0.0011)	[0.097, 0.101]	152,398
2018	0.1020	(0.0010)	[0.100, 0.104]	151,572
2019	0.1021	(0.0011)	[0.100, 0.104]	150,773
2022	0.0940	(0.0010)	[0.092, 0.096]	130,659
2023	0.0932	(0.0010)	[0.091, 0.095]	133,978
2024	0.0929	(0.0010)	[0.091, 0.095]	137,271
2025	0.0976	(0.0010)	[0.096, 0.100]	150,476

Design-based standard errors in parentheses, computed from the PNAD Contínua strata, primary sampling units and first-visit weights (V1032). Dependent variable: log real hourly earnings in the main job (habitual monthly earnings at 2025 prices divided by 52/12 times weekly hours). Sample: occupied persons aged 25-64 with positive earnings and 1-120 weekly hours. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Coefficient on years of schooling in the linear specification with state fixed effects, one regression per year. No first-visit microdata were released for 2020 and 2021.

Table 3: Mincer regressions, pooled 2012-2025

	(1) Linear	(2) Levels	(3) Interactions
Years of schooling	0.0976***		0.1080***
	(0.0005)		(0.0008)
Fundamental incomplete		0.2455***	
		(0.0050)	
Fundamental complete		0.4248***	
		(0.0055)	
Médio incomplete		0.4794***	
		(0.0058)	
Médio complete		0.6340***	
		(0.0054)	
Superior incomplete		0.9275***	
		(0.0068)	
Superior complete		1.4470***	
		(0.0072)	

	(1) Linear	(2) Levels	(3) Interactions
Schooling × Women			0.0092***
			(0.0004)
Schooling × Black/Brown			-0.0251***
			(0.0007)
Schooling × Indigenous			-0.0292***
			(0.0039)
Women	-0.2510***	-0.2639***	-0.3518***
	(0.0016)	(0.0015)	(0.0042)
Black/Brown	-0.1532***	-0.1177***	0.1208***
	(0.0025)	(0.0021)	(0.0062)
Indigenous	-0.1643***	-0.1384***	0.1511***
	(0.0151)	(0.0150)	(0.0386)
Age	0.0463***	0.0414***	0.0461***
	(0.0007)	(0.0007)	(0.0007)
Age squared	-0.0004***	-0.0004***	-0.0004***
	(0.0000)	(0.0000)	(0.0000)
Urban	0.2020***	0.2592***	0.2106***
	(0.0036)	(0.0037)	(0.0036)
Observations	1,773,567	1,773,567	1,773,567
State and year fixed effects	Yes	Yes	Yes

Design-based standard errors in parentheses, computed from the PNAD Contínua strata, primary sampling units and first-visit weights (V1032). Dependent variable: log real hourly earnings in the main job (habitual monthly earnings at 2025 prices divided by 52/12 times weekly hours). Sample: occupied persons aged 25-64 with positive earnings and 1-120 weekly hours. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Education levels are relative to no schooling; interactions are relative to men and to White/Asian workers.

The controls behave as expected and are stable across specifications. Conditional on schooling, age, area and state, women earn 25% log points less per hour than men and Black and Brown workers 15% log points less than White and Asian workers; urban workers earn 20% log points more than rural workers. The age profile peaks at 59 years.

5.2 Premia by completed level

Column (2) of Table 3 replaces years of schooling by the completed level, and Table 4 and Figure 2 show how the level premia moved. The profile is strongly convex. Relative to workers with no schooling, a completed fundamental education (nine years) is associated with 0.42 log points more per hour, or 4.7% per year of schooling; a completed secondary education adds 0.21 log points over three more years, or 7.0% per year; and a completed degree adds 0.81 log points over four more years, or 20.3% per year. Completion matters beyond the years it takes: the jump from an incomplete to a complete fundamental education is 0.18 log points, from

an incomplete to a complete secondary education 0.15, and from an incomplete to a complete degree 0.52. These sheepskin effects are the Brazilian counterpart of the pattern evidence that Roodman (2022) finds most robust in the Indonesian data, where effects concentrate at the grade levels where they are expected.

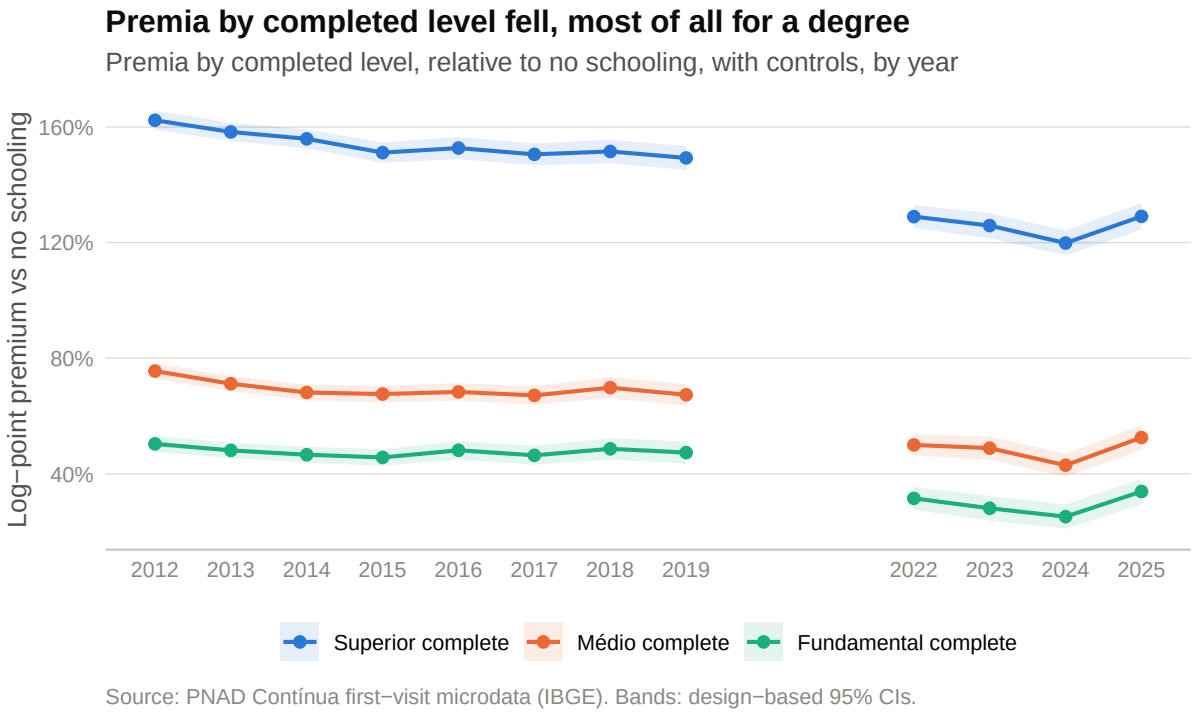


Figure 2: Earnings premia by completed level of education, relative to no schooling, with the same controls as the linear specification; one regression per year. Shaded bands are design-based 95 percent confidence intervals.

Table 4: Earnings premia by completed level of education, relative to no schooling

	2012	2019	2025	Pooled
Fundamental incomplete	0.315***	0.286***	0.187***	0.246***
	(0.013)	(0.018)	(0.022)	(0.005)
Fundamental complete	0.503***	0.473***	0.339***	0.425***
	(0.014)	(0.020)	(0.023)	(0.005)
Médio incomplete	0.579***	0.508***	0.378***	0.479***
	(0.016)	(0.021)	(0.023)	(0.006)
Médio complete	0.756***	0.673***	0.526***	0.634***
	(0.014)	(0.019)	(0.022)	(0.005)
Superior incomplete	1.133***	0.956***	0.792***	0.928***
	(0.018)	(0.023)	(0.025)	(0.007)
Superior complete	1.623***	1.493***	1.291***	1.447***
	(0.017)	(0.021)	(0.023)	(0.007)
Observations	146,662	150,773	150,476	1,773,567

	2012	2019	2025	Pooled

Design-based standard errors in parentheses, computed from the PNAD Contínua strata, primary sampling units and first-visit weights (V1032). Dependent variable: log real hourly earnings in the main job (habitual monthly earnings at 2025 prices divided by 52/12 times weekly hours). Sample: occupied persons aged 25-64 with positive earnings and 1-120 weekly hours. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Controls: age, age squared, sex, race, urban, state fixed effects (and year fixed effects in the pooled column).

Every level premium fell over the period. Relative to no schooling, the premium for a completed degree declined from 1.62 log points in 2012 to 1.49 in 2019 and 1.29 in 2025; the premium for a completed secondary education from 0.76 to 0.53; and the premium for a completed fundamental education from 0.50 to 0.34. The decline is proportionally similar across levels, which points to two forces rather than to a collapse of the tertiary premium alone: the reference group of workers with no schooling shrank and aged while real minimum wages rose through the 2010s, lifting the bottom of the distribution, and the supply of workers at every completed level expanded. How can every level premium fall while the single Mincer coefficient barely moves? The linear coefficient is a weighted average of the local slopes of the profile, and as the workforce moved up a convex profile, with mean schooling rising by almost two years, more weight fell on the steep segment between secondary and tertiary education. The arithmetic is consistent with the data, but it is an interpretation, and the paper does not decompose the change formally.

5.3 Sex and race

Column (3) of Table 3 interacts schooling with sex and race, and Figure 3 traces the resulting group-specific gradients year by year. Pooled over all years, the return to a year of schooling is 10.8% for White and Asian men, 11.7% for White and Asian women, 8.3% for Black and Brown men and 9.2% for Black and Brown women; each is estimated with a standard error below 0.08% points. Two conditional gaps follow. The gender gap narrows with schooling: at nine years of schooling women earn 27% log points less than men with the same characteristics, at sixteen years 20% log points less. The racial gap widens with schooling: it is 10% log points at nine years, 18% at twelve and 28% at sixteen. The ordering of the four gradients is the same in every survey year, and the post-pandemic dip is visible for all groups.

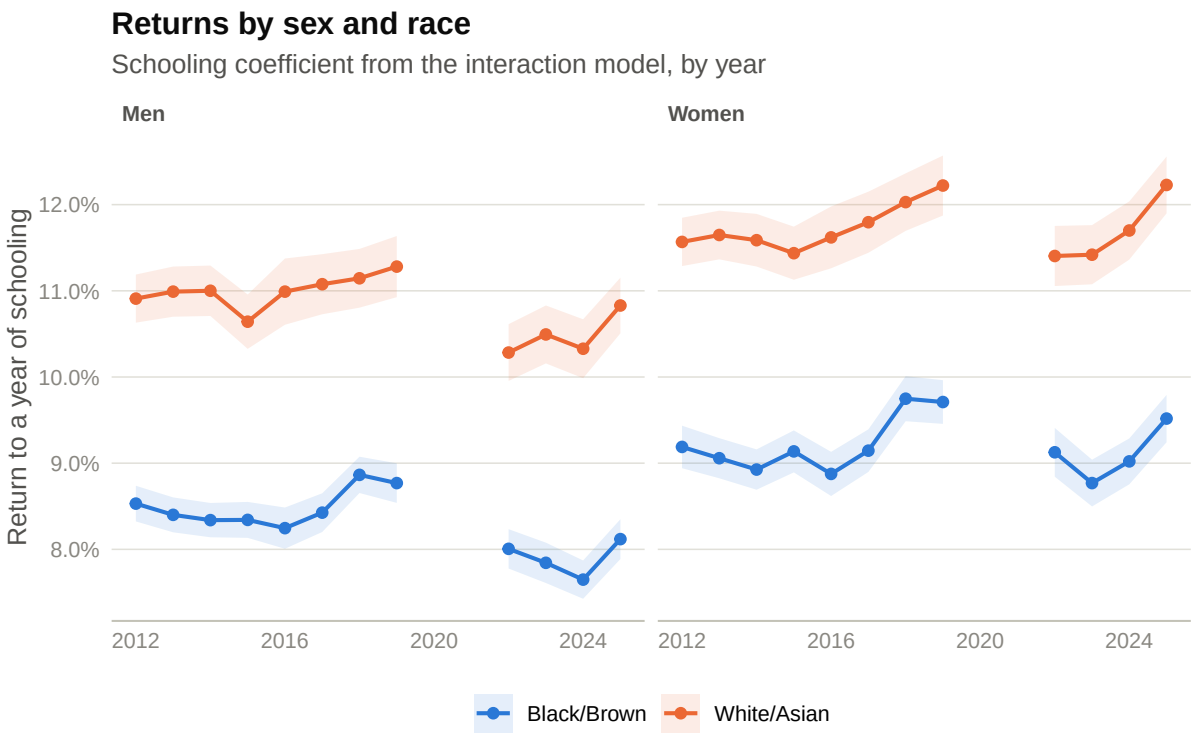


Figure 3: Return to a year of schooling by sex and race. Group-specific schooling coefficients from the interaction specification, one regression per year, with design-based 95 percent confidence intervals.

These are conditional associations. Selection into employment differs by sex and race, and the earnings of those who work are observed only for them; a higher gradient for women can reflect stronger positive selection of less-educated women out of paid work as much as a higher pay-off to schooling. The paper reports the gradients because they are precisely measured facts about who earns what; it does not read them as returns in the causal sense.

5.4 Regions

Table 5 and Figure 4 re-estimate the pooled linear specification within each macro-region. The gradient is steepest in the Nordeste, 10.2%, followed by the Sudeste, 9.9%, the Norte, 9.4%, the Centro-Oeste, 9.2%, and the Sul, 8.9%. The ordering matches the ranking of regions by scarcity of educated workers and is the pattern Duflo reports across Indonesian regions, where returns were higher where initial education was lower (her Table 6).

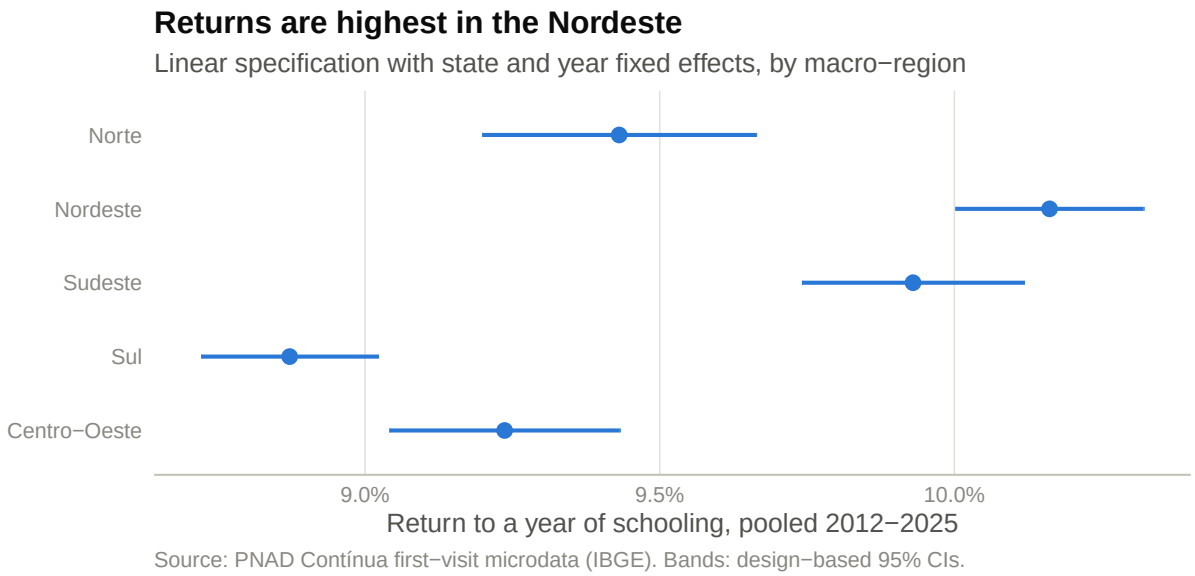


Figure 4: Return to a year of schooling by macro-region, pooled 2012–2025, linear specification with state and year fixed effects estimated separately by region, with design-based 95 percent confidence intervals.

Table 5: Return to a year of schooling by macro-region, pooled 2012-2025

Region	Estimate	SE	95% CI	Observations
Norte	0.0943	(0.0012)	[0.092, 0.097]	227,543
Nordeste	0.1016	(0.0008)	[0.100, 0.103]	505,528
Sudeste	0.0993	(0.0010)	[0.097, 0.101]	506,295
Sul	0.0887	(0.0008)	[0.087, 0.090]	336,062
Centro-Oeste	0.0924	(0.0010)	[0.090, 0.094]	198,139

Design-based standard errors in parentheses, computed from the PNAD Contínua strata, primary sampling units and first-visit weights (V1032). Dependent variable: log real hourly earnings in the main job (habitual monthly earnings at 2025 prices divided by 52/12 times weekly hours). Sample: occupied persons aged 25-64 with positive earnings and 1-120 weekly hours. + $p < 0.1$, * $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$. Linear specification with state and year fixed effects, estimated separately by region.

6 Robustness

Table 6 re-estimates the pooled linear coefficient under six sample and measurement variants and two inference shortcuts. The point estimate is 9.76% at baseline; 10.02% with effective instead of habitual earnings; 10.19% for formal workers only; 9.89% for wage employees only, the sample closest to Duflo's; 9.54% when workers aged 18–24 are included; and 9.24% when the 0.5 percent tails of log earnings are dropped. The range of 9.2% to 10.2% is narrow relative to the differences across groups and levels reported above, and the direction of each change is as expected: formal and better-measured earnings carry a slightly steeper gradient, and young

workers, whose earnings are still rising, a flatter one.

Table 6: Robustness of the pooled linear estimate

Variant	Years of schooling	SE	95% CI	Observations
(1) Baseline	0.0976	(0.0005)	[0.0967, 0.0986]	1,773,567
(2) Effective earnings	0.1002	(0.0005)	[0.0992, 0.1012]	1,728,784
(3) Formal only	0.1019	(0.0006)	[0.1007, 0.1030]	1,044,622
(4) Wage employees	0.0989	(0.0005)	[0.0979, 0.0999]	1,174,059
(5) Ages 18-64	0.0954	(0.0005)	[0.0945, 0.0963]	2,056,891
(6) Trimmed 0.5/99.5	0.0924	(0.0004)	[0.0916, 0.0933]	1,755,831
(7) Unweighted OLS, HC1	0.0936	(0.0001)	[0.0933, 0.0939]	1,773,567
(8) Weighted OLS, cluster UPA	0.0976	(0.0005)	[0.0966, 0.0987]	1,773,567

Coefficient on years of schooling in the linear specification with state and year fixed effects, pooled 2012-2025. Rows (1)-(6) use design-based inference (PNAD Contínua strata, PSUs and first-visit weights); (7) ignores weights and design and uses HC1 standard errors; (8) uses the weights with standard errors clustered by UPA but ignores strata. (2) uses effective instead of habitual earnings; (3) keeps signed-card or statutory employees and contributing self-employed; (4) keeps wage employees; (5) widens the age range; (6) drops the 0.5% tails of log earnings.

The inference comparison is the more instructive part of the table. Unweighted OLS with HC1 standard errors, which ignores the sampling design, moves the point estimate to 9.36% and reports a standard error of 0.00015, 3.4 times smaller than the design-based 0.00049. The difference comes from the clustering of households within primary sampling units, which the robust standard error treats as independent observations. Weighted OLS with standard errors clustered by primary sampling unit recovers the design-based point estimate exactly and a standard error of 0.00052, slightly above the design-based one because it ignores the gain from stratification. For a coefficient this precisely estimated the practical consequence is small, but the same factor of three applies to every group and year comparison in the paper, and it is the kind of choice that Roodman (2022) found undocumented in the original study.

7 Limitations

The strongest objection is that none of these coefficients is the causal effect of schooling on earnings. Workers with more schooling differ in ability, family background and the quality of the schools they attended, all of which raise earnings directly, and the sample contains only

people who work and report earnings. The gradient can therefore overstate or understate the causal return, and the design-based standard errors, however carefully computed, say nothing about that bias. Duflo's Indonesian estimates were built to address exactly this problem, and even they turn out to be weakly identified under reanalysis. The paper's defence is not that the problem is small but that it is stated: the estimates are reported as what they are, precisely measured conditional associations in a representative sample.

Several measurement issues matter for the magnitudes. Hourly earnings are constructed from habitual monthly earnings and habitual weekly hours, both reported by the respondent or by another household member, so measurement error in hours enters the dependent variable directly; the trimmed and effective-earnings variants bound the influence of outliers but not of systematic misreporting. Years of schooling are IBGE's standardised measure and do not capture school quality, which varies enormously across Brazilian states and between public and private schools. The sample includes the self-employed and employers, whose earnings mix returns to labour and to capital; the wage-employee variant shows that this choice moves the pooled coefficient by little.

Two features of the data limit comparisons over time. No first-visit microdata exist for 2020 and 2021, so the post-pandemic decline cannot be dated within those years. For 2022, the current microdata file gives mean labour income 3.3 percent above the values IBGE published for that year, while every other year matches within 0.01 percent; the paper uses the microdata as released and records the discrepancy. More generally, all files are the 2025 revision of the series, re-weighted to post-Census population projections, and the published aggregates may be revised again.

Finally, the PNAD Contínua records the state of residence, not of birth or of schooling. Inter-state migration is selective, so the regional gradients describe where educated workers earn, not where they were educated. A design in the spirit of Duflo's, which needs region of birth and a documented allocation rule, is beyond this survey and is the natural next paper in the series.

8 Conclusion

This paper set out to estimate, for Brazil and with inference that respects the survey design, the OLS Mincer benchmark that Duflo (2001) reports before her instrumental-variables estimates. In 12 survey years between 2012 and 2025, a year of schooling is associated with about 10% higher hourly earnings among occupied workers aged 25–64, with year-by-year estimates between 9.3% and 10.2% and the lowest values in 2022–2024. The single coefficient conceals a convex profile in which a completed degree is worth 1.45 log points relative to no schooling and the step from secondary to tertiary education carries an implied 20% per year, and it conceals a decline in every level premium over the period that is consistent with a workforce moving up that profile as schooling expanded. The gradient is steeper for women than for men, flatter for Black and Brown workers, and steepest in the Nordeste.

The exercise also shows what careful descriptive work requires. Reproducing IBGE's published aggregates from the microdata caught the exact definitions of the official series and exposed a

discrepancy in one year; stating the inference choices showed that the common shortcut of robust standard errors on unweighted data understates uncertainty by a factor of three. Those are the two lessons of the reanalysis of Duflo's paper, applied before any causal claim is attempted.

The causal question remains open for Brazil. Answering it in Duflo's spirit requires a program with a documented allocation rule and data on region of birth, which points to the 1998 FUNDEF school-finance reform and the 2010 or 2022 Censuses rather than to the PNAD Contínua. The estimates in this paper are the benchmark against which such a design should be judged.

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Appendix

Table 7: Sample construction: unweighted counts at each filter step

Year	All persons	Aged 25-64	Occupied	Positive earnings	Hours 1-120	Analysis sample
2012	446,445	224,410	153,037	146,662	146,662	146,662
2013	461,033	233,512	159,719	152,993	152,993	152,993
2014	465,038	239,141	163,589	156,830	156,830	156,830
2015	459,273	238,095	161,463	155,348	155,348	155,348
2016	459,718	239,934	159,229	154,616	154,616	154,607
2017	457,992	239,895	156,825	152,414	152,414	152,398
2018	452,654	238,772	155,788	151,588	151,588	151,572
2019	443,790	234,851	154,705	150,793	150,793	150,773
2022	376,821	201,191	133,465	130,693	130,693	130,659
2023	383,049	204,904	136,580	134,015	134,015	133,978
2024	381,815	205,223	139,788	137,298	137,298	137,271
2025	408,364	220,959	152,873	150,503	150,503	150,476

First-visit annual microdata, PNAD Contínua. Each column applies one more filter to the column to its left: aged 25-64; occupied in the reference week; positive habitual earnings in the main job; 1-120 weekly hours in the main job; schooling and race reported.

Table 8: Official aggregates reproduced from the survey design

Check	Estimate	Official	Rel. diff.	Status
2012 mean real labour income, 14+ occupied, habitual (R$ 2025)	3082.610	3083.000	0.013%	pass
2012 mean real labour income, 14+ occupied, effective (R$ 2025)	3114.480	3114.000	0.015%	pass
2013 mean real labour income, 14+ occupied, habitual (R$ 2025)	3184.870	3185.000	0.004%	pass
2013 mean real labour income, 14+ occupied, effective (R$ 2025)	3210.320	3210.000	0.010%	pass
2014 mean real labour income, 14+ occupied, habitual (R$ 2025)	3279.930	3280.000	0.002%	pass
2014 mean real labour income, 14+ occupied, effective (R$ 2025)	3303.460	3303.000	0.014%	pass

Check	Estimate	Official	Rel. diff.	Status
2015 mean real labour income, 14+ occupied, habitual (R$ 2025)	3151.410	3151.000	0.013%	pass
2015 mean real labour income, 14+ occupied, effective (R$ 2025)	3194.440	3194.000	0.014%	pass
2016 mean real labour income, 14+ occupied, habitual (R$ 2025)	3163.870	3164.000	0.004%	pass
2016 mean real labour income, 14+ occupied, effective (R$ 2025)	3295.310	3295.000	0.010%	pass
2016 mean years of schooling, 25+, Total	9.120	9.100	0.210%	pass
2016 mean years of schooling, 25+, Homens	8.930	8.900	0.327%	pass
2016 mean years of schooling, 25+, Mulheres	9.290	9.200	1.002%	pass
2017 mean real labour income, 14+ occupied, habitual (R$ 2025)	3147.430	3147.000	0.014%	pass
2017 mean real labour income, 14+ occupied, effective (R$ 2025)	3254.120	3254.000	0.004%	pass
2017 mean years of schooling, 25+, Total	9.240	9.200	0.466%	pass
2017 mean years of schooling, 25+, Homens	9.050	9.000	0.575%	pass
2017 mean years of schooling, 25+, Mulheres	9.420	9.400	0.179%	pass
2018 mean real labour income, 14+ occupied, habitual (R$ 2025)	3216.720	3217.000	0.009%	pass
2018 mean real labour income, 14+ occupied, effective (R$ 2025)	3343.950	3344.000	0.002%	pass
2018 mean years of schooling, 25+, Total	9.430	9.400	0.287%	pass
2018 mean years of schooling, 25+, Homens	9.220	9.200	0.211%	pass
2018 mean years of schooling, 25+, Mulheres	9.620	9.600	0.166%	pass
2019 mean real labour income, 14+ occupied, habitual (R$ 2025)	3204.130	3204.000	0.004%	pass
2019 mean real labour income, 14+ occupied, effective (R$ 2025)	3325.570	3326.000	0.013%	pass
2019 mean years of schooling, 25+, Total	9.530	9.500	0.298%	pass
2019 mean years of schooling, 25+, Homens	9.310	9.300	0.054%	pass
2019 mean years of schooling, 25+, Mulheres	9.730	9.700	0.324%	pass
2022 mean real labour income, 14+ occupied, habitual (R$ 2025)	3134.070	3032.000	3.366%	documented exception
2022 mean real labour income, 14+ occupied, effective (R$ 2025)	3254.660	3150.000	3.323%	documented exception

Check	Estimate	Official	Rel. diff.	Status
2022 mean years of schooling, 25+, Total	9.830	9.800	0.263%	pass
2022 mean years of schooling, 25+, Homens	9.610	9.600	0.088%	pass
2022 mean years of schooling, 25+, Mulheres	10.020	10.000	0.229%	pass
2023 mean real labour income, 14+ occupied, habitual (R$ 2025)	3249.040	3249.000	0.001%	pass
2023 mean real labour income, 14+ occupied, effective (R$ 2025)	3371.440	3371.000	0.013%	pass
2023 mean years of schooling, 25+, Total	9.930	9.900	0.326%	pass
2023 mean years of schooling, 25+, Homens	9.730	9.700	0.322%	pass
2023 mean years of schooling, 25+, Mulheres	10.110	10.100	0.145%	pass
2024 mean real labour income, 14+ occupied, habitual (R$ 2025)	3368.090	3368.000	0.003%	pass
2024 mean real labour income, 14+ occupied, effective (R$ 2025)	3492.080	3492.000	0.002%	pass
2024 mean years of schooling, 25+, Total	10.050	10.100	0.482%	pass
2024 mean years of schooling, 25+, Homens	9.810	9.900	0.878%	pass
2024 mean years of schooling, 25+, Mulheres	10.270	10.300	0.316%	pass
2025 mean real labour income, 14+ occupied, habitual (R$ 2025)	3559.530	3560.000	0.013%	pass
2025 mean real labour income, 14+ occupied, effective (R$ 2025)	3694.130	3694.000	0.003%	pass
2025 mean years of schooling, 25+, Total	10.240	10.200	0.386%	pass
2025 mean years of schooling, 25+, Homens	10.010	10.000	0.139%	pass
2025 mean years of schooling, 25+, Mulheres	10.440	10.400	0.422%	pass

Estimate: design-based mean from the first-visit microdata (weights V1032, strata, PSUs). Official: IBGE/SIDRA tables 7445 (labour income, same microdata and deflator) and 7126 (years of schooling, second-quarter supplement, one decimal).